# DATA MODELING PROJECT

**Problem Statement:**
A global retail organization operates across multiple source systems, generating high volumes of transactional data through both CRM and ERP platforms that are currently stored in inconsistent formats and disjointed silos. Business teams face significant challenges in answering operational questions due to fragmented data, calculation errors in sales records, and a lack of standardized categorical values like gender, marital status, and country codes. To resolve this, you are required to build a centralized backend data platform that ingests raw sales and transaction data, cleans and standardizes it, and organizes it into an analytics-ready Star Schema using a Medallion Architecture. The focus is on designing scalable schemas and optimized SQL transformations—including deduplication, historical tracking, and surrogate key implementation—to enable accurate and timely retail insights for reporting and downstream analytics.

**Project Description:**
This project focuses on designing and implementing a modern data engineering solution for a retail business. The goal is to transform raw, unstructured, and inconsistent transactional data into a clean, reliable, and analytics-ready format. The system will follow the Medallion Architecture:

**1. Bronze Layer:** Ingest raw data from CRM and ERP systems without transformations. This layer preserves the original data for auditing and debugging purposes.

**2. Silver Layer:** Perform data cleaning and standardization. This includes handling missing values, removing duplicates, fixing inconsistent formats, and standardizing categorical fields such as gender, country, and status codes.

**3. Gold Layer:** Transform the cleaned data into a Star Schema. This includes fact tables (sales, transactions) and dimension tables (customers, products, time, geography). Implement surrogate keys and ensure proper relationships for analytical queries.

**Key Features:**
- Data deduplication and validation
- Slowly Changing Dimensions (SCD) for historical tracking
- Optimized SQL transformations for performance
- Scalable schema design for large datasets
- Support for BI tools and reporting dashboards

**Expected Outcome:**
A robust and scalable data platform that enables business users to generate insights such as sales trends, customer behavior, and regional performance efficiently and accurately.